

Statistical Methods in the MINERvA-OmniFold 2D Uncertainty Quantification Campaign

For the motivated undergraduate

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Abstract

This note explains every statistical method used in the uncertainty quantification (UQ) campaign for the 2D MINERvA-OmniFold $d^2\sigma/(dp_T dp_{\parallel})$ analysis, in enough depth that a reader who is comfortable with detector physics, OmniFold, MLE, gaussians, χ^2 , and p -tests — but new to bootstrap, multivariate covariance, pulls, frequentist coverage, and shape-only χ^2 — can answer any question about how the analysis quantifies uncertainty. Each procedure section gives (a) what the procedure estimates, (b) the recipe, (c) the math (with one-line derivations where useful), (d) a pointer to the implementation, and (e) a worked example using a current reported number, so the formulas land in the same place as the headline results. It also documents *how MINERvA itself* builds the published covariance (§3.4), and the rank-deficiency obstruction to an ours-only χ^2 together with the open questions we are taking to MINERvA collaborators (§5). This note consolidates and supersedes the earlier `advisor_memo_cov_rank.md` and `reference/minerva_systematics_sources.md`.

Contents

1	Orientation	1
2	Statistical prerequisites you may not have yet	2
2.1	Sample mean, sample variance, Bessel's correction	2
2.2	Covariance matrices and the multivariate Gaussian	3
2.3	Inverting a covariance: Cholesky, ridge jitter, pseudo-inverse	3
2.4	The bootstrap, in one page	4
2.5	Frequentist coverage	4
2.6	Pull and standardised residual	5
2.7	Jacobian propagation of covariance	5
3	Variance-source procedures	5
3.1	ML stochasticity: the seedscan	5
3.2	Statistical uncertainty: the Poisson bootstrap	6

3.3	Systematic uncertainty: universes	8
3.4	How MINERvA builds its published covariance	10
3.5	Block-sum combination	11
4	Comparing to the paper	12
4.1	The 205-bin selection	12
4.2	Covariance-aware χ^2/ndf	12
4.3	Pull distribution	13
4.4	Shape-only χ^2 with Jacobian propagation	13
5	Ill-conditioning, rank deficiency, and open questions	14
6	Anatomy of the paper-covariance tension	16
7	Calibration: coverage toys	18
8	Closure tests: detecting bias	19
8.1	Truth-reweight closure	19
8.2	Hidden-variable closure	20
8.3	Alternative-model closure	21
9	How the pieces fit together	21
A	Symbol table	23
B	Reproducing the numbers	23
C	Multinomial $\rightarrow$ independent Poisson bootstrap limit	24
D	The $\pm 1\sigma$ pair covariance estimator: MAT vs legacy	24

1 Orientation

OmniFold returns, in this analysis, a set of weighted MC events whose binning over $(p_T, p_{\parallel})$ yields a $14 \times 16 = 224$ -cell double-differential cross section x_i . We compare against the published MINERvA result (arXiv:2106.16210), which reports 205 of those bins ($x_i^{\text{paper}} > 0$); the remaining 19 are acceptance- suppressed and excluded everywhere. “Uncertainty,” for us, is a 205-dimensional Gaussian-approximated covariance matrix $\mathbf{C}$ over those bins. Building $\mathbf{C}$ and using it correctly is the entire job of the UQ machinery.

The campaign factorises uncertainty into three statistically independent variance sources and adds four diagnostic tests:

Component	Estimates	Section
ML seedscan	Stochasticity of the GBDT optimiser	§3.1
Poisson bootstrap	Data + MC statistical variance	§3.2
Systematic universes	Model uncertainty (flux, GENIE, detector)	§3.3
Block-sum combination	Independent quadrature of the above	§3.5
χ^2/ndf vs paper	Goodness-of-fit, headline number	§4.2
Pull distribution	Outlier diagnostic	§4.3
Shape-only χ^2	Disentangle shape from absolute scale	§4.4
Coverage toys	Calibration check on the bootstrap band	§7
Closure tests	Detect bias (not variance) in the unfold	§8

The three variance sources combine in quadrature. As per-bin median relative spreads over the 205 reported bins,

$$\underbrace{0.166\%}_{\text{ML}} \oplus \underbrace{0.564\%}_{\text{stat}} \oplus \underbrace{4.783\%}_{\text{syst}} \longrightarrow \underbrace{4.822\%}_{\text{combined}}, \quad \text{paper: } 6.86\%, \quad (1)$$

where $\oplus$ denotes per-bin quadrature followed by taking the median (medians do not literally add, so this is a decomposition, not an identity). Systematics dominate; data-statistics are a half-percent effect; ML stochasticity is a 0.17% noise floor. Everything below uses one 205-dimensional vector $\mathbf{x}$ for the bin contents and one 205×205 matrix $\mathbf{C}$ for their covariance.

2 Statistical prerequisites you may not have yet

You already know that a Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ has likelihood $\mathcal{L}(\mu, \sigma; x) \propto \exp[-(x - \mu)^2/(2\sigma^2)]$, and that maximum-likelihood estimation (MLE) of μ from N independent samples gives the sample mean. The list below upgrades that to the language used in the rest of this note.

2.1 Sample mean, sample variance, Bessel’s correction

Given N independent draws $x_1, \dots, x_N$ from a distribution with true mean μ and true variance σ^2 , the obvious estimators are

$$\bar{x} = \frac{1}{N} \sum_{k=1}^N x_k, \quad \hat{\sigma}_N^2 = \frac{1}{N} \sum_{k=1}^N (x_k - \bar{x})^2. \quad (2)$$

The mean is unbiased: $\mathbb{E}[\bar{x}] = \mu$. The variance estimator is *biased*: $\mathbb{E}[\hat{\sigma}_N^2] = \frac{N-1}{N} \sigma^2$. Subtracting $\bar{x}$ instead of μ removes one “degree of freedom” from the residuals. Dividing by $N - 1$ instead of N restores unbiasedness:

$$s^2 = \frac{1}{N-1} \sum_{k=1}^N (x_k - \bar{x})^2, \quad \mathbb{E}[s^2] = \sigma^2. \quad (3)$$

This is Bessel’s correction. `ddof=1` in NumPy is exactly this choice. **Note for §3.3:** MINERvA’s MAT framework uses the *biased* $1/N$ estimator for its per-band covariance, not Bessel’s $1/(N - 1)$, and we match that convention in the systematic rollup (§3.4). The bootstrap and seedscan rollups, which estimate variances of our own Monte Carlos, use the unbiased $1/(N - 1)$.

2.2 Covariance matrices and the multivariate Gaussian

For n random variables $X_1, \dots, X_n$, the covariance matrix $\mathbf{C}$ has entries

$$C_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]. \quad (4)$$

Diagonals $C_{ii} = \sigma_i^2$ are the per-bin variances; off-diagonals C_{ij} encode correlations. The correlation matrix is $\rho_{ij} = C_{ij}/(\sigma_i\sigma_j) \in [-1, 1]$.

The natural n -dimensional generalisation of the Gaussian is

$$\mathcal{L}(\boldsymbol{\mu}, \mathbf{C}; \mathbf{x}) \propto \frac{1}{\sqrt{\det \mathbf{C}}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right]. \quad (5)$$

Taking $-2 \ln \mathcal{L}$ and dropping constants gives, modulo the log-determinant,

$$\chi^2(\mathbf{x}; \boldsymbol{\mu}, \mathbf{C}) = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu}). \quad (6)$$

This is the formula you already know from 1D χ^2 tests, generalised: $1/\sigma^2$ becomes $\mathbf{C}^{-1}$, and correlated deviations get reweighted by the inverse covariance. In the special case $\mathbf{C} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, Eq. (6) reduces to $\sum_i (x_i - \mu_i)^2/\sigma_i^2$. Under the null hypothesis $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$, this χ^2 follows a χ_n^2 distribution with mean n , hence *reduced* $\chi^2/n \approx 1$ when the model is correct.

2.3 Inverting a covariance: Cholesky, ridge jitter, pseudo-inverse

Equation (6) requires inverting $\mathbf{C}$. Three practical issues come up in this analysis.

Positive-definiteness check (Cholesky). A matrix $\mathbf{C}$ is positive-definite (PD) iff there is a lower-triangular $\mathbf{L}$ with $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ (the Cholesky factorisation). If Cholesky succeeds, $\mathbf{C}$ is invertible and Eq. (6) is well-posed.

Ridge jitter. Numerical noise can leave a PD matrix barely non-PD. The standard fix is to add an isotropic ridge $\varepsilon \mathbf{I}$. The `uq/analyze_uq.py` script does exactly this:

$$\varepsilon = 10^{-12} \frac{\text{Tr} \mathbf{C}}{n}, \quad (7)$$

where the trace sets the natural scale of the eigenvalues. Twelve orders of magnitude below the mean diagonal is far below any real signal in $\mathbf{C}$ and safely above floating-point round-off.

Rank deficiency at $N \leq n$. If you estimate $\mathbf{C}$ from N replicas in a space of n bins with $N \leq n$, the resulting sample covariance has rank at most $N - 1$: there are not enough independent “directions” to fill the full $n \times n$ matrix. This was the regime of the Stage-1 bootstrap ($N = 50$ replicas, $n = 205$ bins): Cholesky failed without jitter, and the recorded `jitter` = 1.6e-94 in the run log was a numerical artefact of the rank deficiency rather than a real near-singularity. The Stage-2 bootstrap ($N = 300 > 205$) is full-rank. The *systematic* covariance suffers a more stubborn version of this problem that survives at any universe count — see §5.

Pseudo-inverse. For the χ^2 comparison against the paper covariance, `compare_to_paper_fullcov.py` replaces $\mathbf{C}^{-1}$ with the Moore–Penrose pseudo-inverse $\mathbf{C}^+$, computed via SVD. Writing $\mathbf{C} = \mathbf{U}\Sigma\mathbf{V}^\top$ with $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$,

$$\mathbf{C}^+ = \mathbf{V} \Sigma^+ \mathbf{U}^\top, \quad \Sigma_{ii}^+ = \begin{cases} 1/\sigma_i & \sigma_i > \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (8)$$

with a small threshold τ (NumPy’s default is $\tau = n \cdot \sigma_{\max} \cdot \varepsilon_{\text{machine}}$). This silently drops singular modes that are numerically indistinguishable from zero, replacing them with 0 in the inverse rather than ∞ . The result is a stable quadratic form even when $\mathbf{C}$ has zero or near-zero eigenvalues.

2.4 The bootstrap, in one page

The bootstrap (Efron 1979) estimates the variance of a statistic $T(\mathbf{x})$ by resampling the data themselves rather than re-running the experiment. The non-parametric form draws N samples with replacement from the original N events; some events appear once, some twice, some not at all. The number of times event i appears, n_i , is multinomial with $\sum n_i = N$ and $\mathbb{E}[n_i] = 1$.

In the large- N limit, the n_i become independent and each is approximately Poisson(1). Equivalently:

$b_i \sim \text{Poisson}(1) \text{ i.i.d. is the per-event multiplicative weight of a non-parametric bootstrap.}$

(9)

For a weighted analysis you replace event weight w_i by $w'_i = w_i b_i$. The expected sum of weights is unchanged ($\mathbb{E}[\sum w'_i] = \sum w_i$), but each replica has fluctuations of order $\sqrt{N}$, reproducing the Poisson statistics of the original event count. Appendix C shows the multinomial $\rightarrow$ independent-Poisson limit explicitly.

2.5 Frequentist coverage

If you claim a 1-sigma uncertainty band $\hat{\mu} \pm \hat{\sigma}$ on some parameter μ , the frequentist guarantee is: under repeated experiments, the true μ should fall in the band 68.27% of the time. Coverage is the empirical version of this: build a band, run many synthetic experiments where you *know* the truth, and count how often the band covers it. A well-calibrated 1-sigma band yields $\sim 68.27\%$ coverage. Significantly higher means conservative (over-coverage); significantly lower means under-coverage.

2.6 Pull and standardised residual

For a measured value x_i with quoted uncertainty σ_i and a reference x_i^{ref} , the *pull* is the standardised residual

$$\text{pull}_i = \frac{x_i - x_i^{\text{ref}}}{\sigma_i}. \quad (10)$$

Under the null hypothesis $x_i \sim \mathcal{N}(x_i^{\text{ref}}, \sigma_i^2)$, the pulls are standard normal: mean 0, RMS 1, roughly symmetric. Outlier bins jump out as $|\text{pull}| \gtrsim 3$, and systematic offsets show up as a non-zero pull mean. RMS $\ll 1$ means the quoted σ is conservative (band too wide for the actual disagreement); RMS $\gg 1$ means the band is too narrow.

2.7 Jacobian propagation of covariance

If $\mathbf{y} = f(\mathbf{x})$ and $\mathbf{x}$ has small Gaussian fluctuations with covariance $\mathbf{C}_x$, then to first order in the fluctuations

$$\mathbf{y} - f(\boldsymbol{\mu}_x) \approx \mathbf{J}(\mathbf{x} - \boldsymbol{\mu}_x), \quad J_{ij} = \frac{\partial f_i}{\partial x_j}, \quad (11)$$

and the covariance of $\mathbf{y}$ is

$$\boxed{\mathbf{C}_y = \mathbf{J} \mathbf{C}_x \mathbf{J}^\top}. \quad (12)$$

This is what the shape-only χ^2 (§4.4) uses: the transformation $\mathbf{x} \mapsto \mathbf{x} / \sum w_j x_j$ is nonlinear, but locally well-approximated by a linear map with a particular $\mathbf{J}$ that we derive in §4.4.

3 Variance-source procedures

Each subsection below follows the same template: *What it estimates* · *Procedure* · *Math* · *Code pointer* · *Worked example*.

3.1 ML stochasticity: the seedscan

What it estimates. The variance in the unfolded cross section due to random choices inside the GBDT (gradient-boosted decision tree) optimiser. OmniFold is deterministic given the input weights, the network architecture, and the random seed; flipping only the seed produces slightly different fits to the per-event reweighting functions, and that spread is the “ML noise floor”.

Procedure. Run $n = 10$ full 5-iteration MEFHC unfolds with everything fixed except the GBDT `random_state`. The driver `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` pins three seeds per trial via `--seed N`:

$$(\text{step-1 classifier}, \text{step-2 classifier}, \text{miss regressor}) = (N, N+1, N+2), \quad N \in \{1, \dots, 10\}. \quad (13)$$

The canonical seedscan uses the `lgbm` (LightGBM) backend, so it is backend-matched to the production unfold; an earlier HistGBT seedscan gave a slightly larger floor and is superseded.

Math. For each reported bin i , build the n -vector $X_i^{(1)}, \dots, X_i^{(n)}$ across trials and compute the unbiased sample standard deviation

$$\sigma_i^{\text{ML}} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (X_i^{(k)} - \bar{X}_i)^2}, \quad \bar{X}_i = \frac{1}{n} \sum_k X_i^{(k)}. \quad (14)$$

The headline diagnostic is the per-bin *relative* spread $\sigma_i^{\text{ML}}/\bar{X}_i$, summarised by its median over the 205 reported bins.

Code pointer. `seedscore/analyze_seedscore.py (rollup); sbatch_unfold_2d_MEFHC_5iter_seedscore.s (driver); covariance output uq/seedscore_lgbm_ml/uq_covariance_ml.root.`

Worked example (STATUS, UQ envelope). With $n = 10$ lgbm trials on the 205-bin set:

- Total- σ relative spread: 0.0055%.
- Per-bin median relative spread: 0.166%.
- Component trace $\sqrt{\text{Tr}\mathbf{C}^{\text{ML}}} = 5.061 \times 10^{-41}$.

The paper’s per-bin total-uncertainty median is 6.86%; the ML noise is therefore $\approx 2.4\%$ of the paper’s quoted uncertainty and far subdominant to data-stat and systematics. Going from $n = 4$ to $n = 10$ trials moved the headline numbers within rounding, so the seedscore envelope is converged.

3.2 Statistical uncertainty: the Poisson bootstrap

What it estimates. The combined data-statistical and MC-statistical variance, propagated *coherently* through the full OmniFold pipeline. Coherent means: if some data event happens to be “up-weighted” in a replica, the same event flows through both the step-1 classifier training and the final reweighted-MC cross-section construction; if an MC event is up-weighted, the same up-weight is applied to its reco-tree and truth-tree copies, preserving the response matrix.

Procedure. For each replica $k = 1, \dots, N$:

1. Draw per-event Poisson(1) factors $b_i^{\text{data},(k)}$ for every data event and $b_i^{\text{MC},(k)}$ for every MC event, using two independent sub-RNGs (a data RNG seeded from k , and an MC RNG seeded from $k + 10^7$). The two sources are uncorrelated by construction.
2. Multiply data weights by b^{data} and MC weights (both reco and truth copies of each MC event) by the *same* b^{MC} . This keeps the migration matrix self-consistent.
3. Run OmniFold (5 iterations) with the GBDT seed *pinned*. Pinning the seed isolates the bootstrap signal from the ML noise of §3.1.
4. Save `hXSec2D` (and 1D projections).

Stage-1 used $N = 50$ replicas (rank-deficient on 205 bins); Stage-2 scaled to $N = 300 > 205$ for a full-rank statistical covariance.

Caveat (seed-varying replicas). The available $N = 300$ covariance was produced with replicas that varied *both* `--bootstrap-seed N` and `--seed N`, so it carries a small ML-stochastic contribution on top of the pure Poisson statistics. The drivers are now patched to pass a fixed `--seed 1` with only `--bootstrap-seed N` varying; a pure-Poisson rerun should replace this block when available. The leak is numerically small relative to the systematic block (the ML floor is 0.166% vs the stat 0.564%), but the statistical covariance is not yet purely Poisson.

Math. Flatten each replica’s 14×16 histogram into a 205-vector $\mathbf{X}^{(k)}$ over reported bins (row-major over $(p_T, p_{\parallel})$). The bootstrap covariance is the unbiased sample covariance:

$$(\mathbf{C}^{\text{boot}})_{ij} = \frac{1}{N-1} \sum_{k=1}^N (X_i^{(k)} - \bar{X}_i)(X_j^{(k)} - \bar{X}_j), \quad \bar{X}_i = \frac{1}{N} \sum_k X_i^{(k)}. \quad (15)$$

This is exactly `np.cov(Xrep, rowvar=False, ddof=1)` in `uq/analyze_uq.py`. The rank-deficiency check is the Cholesky-plus-jitter pattern from §2.3.

Why Poisson(1) is the right primitive. A non-parametric bootstrap with replacement assigns event i a multinomial count n_i with $\mathbb{E}[n_i] = 1$, $\text{Var}[n_i] = (N-1)/N \rightarrow 1$, and $\mathbb{E}[\sum n_i] = N$. In the large- N limit the n_i become independent and each is Poisson(1) (Appendix C). For a weighted analysis you multiply $w_i \rightarrow w_i n_i$; since n_i may equal zero, some events are dropped from a given replica, exactly mimicking the “with replacement” resample.

Code pointer. Bootstrap weight draws and seed pinning: `unfold_2d.omnifold.unbinned.py` (`--bootstrap-seed`, `--seed`).

Covariance rollup: `uq/analyze_uq.py`.

Output: `uq/bootstrap_MEFHC_300/uq_covariance_boot300.root`.

Worked example (STATUS, $N = 50$ vs $N = 300$, MEFHC lgbm).

Metric (205 bins)	$N = 50$	$N = 300$
Total σ relative std	0.068%	0.061%
Per-bin median relative spread	0.532%	0.564%
Per-bin p_{84}	1.235%	1.215%

The per-bin median is stable from $N = 50$ to $N = 300$ (0.53% $\rightarrow$ 0.56%); the $N = 300$ block is the one used in the headline because it is full rank ($N > 205$), whereas $N = 50$ was rank-deficient and its Cholesky needed jitter 1.6×10^{-94} . The bootstrap median is $\approx 3.4 \times$ the seedscan median (0.166%), as expected for a sample of $\mathcal{O}(10^6)$ data events. Across datasets the per-bin median scales as $\sqrt{N_{\text{events}}}$: playlist 1A ($\approx N/12$ of MEFHC) gives 1.46%, consistent with $\sqrt{12} \times 0.56\% \approx 1.9\%$ at the same backend.

3.3 Systematic uncertainty: universes

What it estimates. Model uncertainty, by re-running the full unfold under each of several physically-motivated reweightings of the MC. Where the bootstrap shakes the *events*, the universes shake the *model*: a different flux $\rightarrow$ different per-event weights $\rightarrow$ different unfolded cross section.

Procedure. The C++ event loop (`runEventLoopOmniFold.cpp`, env `MNV101_DUMP_UNIVERSES`) dumps, for each MC event, alternative weight branches $w_{B,u}^{\text{truth}}$ and $w_{B,u}^{\text{reco}}$ for every (band B , universe index u) pair. The publication-grade Stage-2 sweep has **187 universes across 44 bands**: 6 *lateral* bands (`BeamAngleX/Y`, `MuonResolution`, `GEANT_{Neutron,Pion,Proton}`, `Muon_Energy_MINERvA`) that require re-running the event loop with shifted detector quantities, plus 38 *vertical* bands that only re-weight. The dominant ensemble is the 100-universe PPFX flux band; almost all other bands are $\pm 1\sigma$ pairs.

Band (examples)	Variation	N_u
Flux	PPFX hadron-production sampling	100
<code>Muon_Energy_MINOS</code>	Muon energy scale, $\pm 1\sigma$	2
<code>Muon_Energy_MINERvA</code>	Muon energy scale (lateral), $\pm 1\sigma$	2
<code>MinosEfficiency</code>	MINOS muon-match efficiency, $\pm 1\sigma$	2
<code>MaRES</code> , <code>MvRES</code> , <code>MaCCQE</code>	GENIE knobs, $\pm 1\sigma$	2 ea.
<code>GEANT_{Neutron,Pion,Proton}</code>	FSI response (lateral), $\pm 1\sigma$	2 ea.

For each (B, u) , `unfold_2d_omnifold_unbinned.py --universe B:u` reweights all MC events (truth and reco) by the matching branch and runs OmniFold with a fixed seed. A separate matched “CV” job runs without `--universe`, with the *same* omnifile, backend, seed, iteration count, and flux, so that ML noise cancels between CV and universe.

Math (MAT-conformant, the current convention). For universe (B, u) , define the per-bin value $\mathbf{X}^{(B,u)}$, a 205-vector over reported bins. MINERvA’s MAT framework (`PlotUtils/MnvVertErrorBand::CalcCovMx`, see §3.4) builds the per-band covariance as the *mean-centered, biased* sample covariance, referenced to the **universe mean** (not the CV) and normalised by $1/N_u$:

$$\mathbf{C}^B = \frac{1}{N_u} \sum_{u=1}^{N_u} (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B) (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B)^\top, \quad \langle \mathbf{X} \rangle_B = \frac{1}{N_u} \sum_u \mathbf{X}^{(B,u)}. \quad (16)$$

This single formula is applied *uniformly* to all $N_u \geq 2$ — there is **no special-case $\pm 1\sigma$ pair formula**. For a $\pm 1\sigma$ pair ($N_u = 2$) it collapses to the rank-1 $\frac{1}{4}(\mathbf{X}_+ - \mathbf{X}_-)(\mathbf{X}_+ - \mathbf{X}_-)^\top$; mean-centering kills the symmetric component, so a one-sided or asymmetric pair correctly contributes near-zero variance rather than the rank-2 contribution the legacy CV-centered formula would give (Appendix D).

Target-nucleon normalization (rank-1 add-on). MINERvA quotes a 1.4% overall target-nucleon normalization uncertainty (Aliaga NIM, arXiv:1305.5199) that lives outside the `MnvVertErrorBand` machinery. We add it as a fully-correlated rank-1 outer product on top of the band sum:

$$\mathbf{C}^{\text{norm}} = (\sigma_N \mathbf{X}^{\text{CV}})(\sigma_N \mathbf{X}^{\text{CV}})^\top, \quad \sigma_N = 0.014, \quad (17)$$

via `analyze_universes.py --add-norm 0.014`.

Total systematic covariance. Bands are assumed independent (different physics sources, different RNGs), so

$$\mathbf{C}^{\text{syst}} = \sum_B \mathbf{C}^B + \mathbf{C}^{\text{norm}}. \quad (18)$$

Independence is a modelling assumption with one known exception — the Bashyal flux $\leftrightarrow$ Muon_Energy_MINOS joint block (§5) — which we do not yet propagate.

Legacy formula (forensic only). The pre-2026-05-28 rollup used a CV-centered pair estimator $\frac{1}{2}(\Delta_+ \Delta_+^\top + \Delta_- \Delta_-^\top)$ with $\Delta = \mathbf{X} - \mathbf{X}^{\text{CV}}$. It agrees with Eq. (16) only for symmetric pairs and over-counts asymmetric ones; moving to the MAT formula dropped the total universe trace by 1.9%. It is reachable via `--legacy-pair-formula` for reproduction of the superseded $\chi^2/\text{ndf} = 1.473$ headline only.

Code pointer. Event-loop dump: `runEventLoopOmniFold.cpp` (env `MNV101_DUMP_UNIVERSES`).
Per-universe unfold: `unfold_2d_omnifold_unbinned.py --universe BAND:IDX`.
Per-band rollup: `uq/analyze_universes.py (--add-norm 0.014)`.
Output: `uq/universe_stage2_MEFHC_full_matcorr/uq_universe_covariance_full_matcorr.root`.

Worked example (STATUS, full matcorr sweep). Top contributing bands, per-bin median relative σ_i/X_i^{CV} over 205 reported bins:

Band	Per-bin median rel. σ
Muon_Energy_MINOS	2.305%
Muon_Energy_MINERvA	1.285%
MinosEfficiency	1.472%
Flux (PPFX, $N_u = 100$)	1.008%
MaRES	0.548%
MvRES	0.377%
MaCCQE	0.357%
Systematic total (44 bands + norm)	4.783% (median); $\sqrt{\text{Tr}\mathbf{C}} = 2.463 \times 10^{-39}$

Grouped by category the medians are: Muon reconstruction 3.366%, Models 2.034%, Flux 1.008%, Hadronic response 0.758%, Statistical 0.563%, Normalization 0.081%. Muon energy scale (MINOS+MINERvA) and the MINOS-efficiency band dominate; these are exactly the bands the Ruterbories paper flags as flux-correlated (§5). (*These are the MAT-*

conformant `matcorr` medians; the pre-2026-05-28 legacy build gave larger per-band values, e.g. `Muon_Energy_MINOS` 2.571%.)

3.4 How MINERvA builds its published covariance

To compare honestly we had to establish what MINERvA actually does — which is *not* spelled out in the paper but lives in the MAT C++ framework. The takeaway: **MAT samples per-band exactly as we do**, using the mean-centered $1/N$ sample covariance, and sums bands element-wise.

What the Ruterbories 2021 paper says (Section V). Three uncertainty categories — flux, detector response, neutrino-interaction model — evaluated by “re-extracting the cross section using modified simulations”, i.e. the universe method, but with *no covariance formula given*. Specifics quoted: flux below 4% across most of phase space thanks to the ν - e -scattering constraint (Valencia, arXiv:1906.00111); 1.4% target-nucleon normalization; in-situ test-beam hadron response with per-particle GEANT inelastic variations; muon energy scale 1% constrained by the low-recoil fit of Bashyal (JINST 16 P08068), with the explicit statement that “*the fit causes the flux and muon energy scale uncertainties to be correlated and that correlation is propagated to the final result*” — the only non-block-diagonal piece the paper admits. The only covariance that appears in equations is V in the model-test χ^2 (Eq. 1, “the measurement covariance matrix”), with 205 degrees of freedom (Table I), and a log-normal χ^2 variant motivated by the highly-correlated flux normalization (Peelle’s Pertinent Puzzle).

What the paper does *not* say. No per-band covariance construction; no $\pm 1\sigma$ pair formula; no band-independence/block-sum statement; nothing about rank, condition number, or regularization of V ; no bootstrap or coverage validation (the paper uses D’Agostini iterative unfolding). All of this is delegated to MAT or simply unstated.

Per-band covariance: `MnvVertErrorBand::CalcCovMx`. Reading the MAT source establishes three facts, encoded in Eq. (16):

- the reference is the **universe mean** $\langle x \rangle_u$, not the central value X_{CV} ;
- the normalization is $1/N$ (biased), not Bessel’s $1/(N - 1)$ — a 1% offset for PPFX ($N = 100$);
- there is **no special case for $N = 2$** — the mean-centered N -universe form is applied uniformly. The “MINERvA-101 $\pm 1\sigma$ pair” formula $\frac{1}{2}(\Delta_+ \Delta_+^\top + \Delta_- \Delta_-^\top)$ is a folk re-derivation that agrees only for CV-symmetric shifts.

We verified our rollup against MAT byte-for-byte: populating a `PlotUtils::MnvH1D` with the same 44 bands / 188 universes and comparing `MnvH1D::GetTotalErrorMatrix` against our `--add-norm 0` rollup gives a max element-wise relative difference of 5.5×10^{-17} (machine ε), with $\sqrt{\text{Tr}}$ matching to seven digits (`uq/verify_matcorr_vs_mnvh1d.py`).

Total covariance: `MnvH1D::GetTotalErrorMatrix`. A straight element-wise sum across bands plus the statistical matrix. Band-independence is assumed; cross-band correlations

exist only if encoded inside a single combined band (e.g. the Bashyal joint fit, handled outside this loop). MAT does *no* ridge / shrinkage / SVD truncation on the published total cov and no special handling of rank deficiency — the published V is full-rank because it sums more bands than we currently propagate (§5).

Upstream references that set each band’s $\pm 1\sigma$ amplitude.

Input	Reference
Flux / PPFX (100 universes)	Aliaga, PRD 94 092005 (2016), arXiv:1607.00704
ν - e flux constraint ($< 4\%$)	Valencia, PRD 100 092001 (2019), arXiv:1906.00111
Muon-E scale $\leftrightarrow$ flux fit	Bashyal, JINST 16 (2021) P08068
Test-beam / GEANT hadron response	Aliaga, NIM A 789 28 (2015), arXiv:1501.06431
GENIE reweight knobs	Andreopoulos, NIM A 614 87 (2010); GENIE manual arXiv:1510.07684
1.4% normalization (detector mass)	Aliaga, NIM A 743 130 (2014), arXiv:1305.5199

3.5 Block-sum combination

What it estimates. The total uncertainty covariance, combining the three statistically independent components above.

Independence assumption. The bootstrap samples *events* (with two distinct sub-RNGs for data and MC). The universes vary *model weights*, with the GBDT seed pinned. The seedscan varies the *GBDT seed*, with all weights fixed. The three Monte Carlos are independent by construction:

$$\mathbf{C}^{\text{tot}} = \mathbf{C}^{\text{ML}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{syst}}. \quad (19)$$

The combined-cov hook in `compare_to_paper_fullcov.py` (`--omnifold-cov ROOT:HIST, repeatable`) adds OmniFold- derived covariances on top of the paper’s `TotalCovariance` through exactly this rule. $\mathbf{C}^{\text{ML}}$ is more than an order of magnitude below the other two; the headline includes it but it is numerically negligible.

Worked example (STATUS, UQ envelope). Component traces and per-bin median relative spreads:

Component	$\sqrt{\text{Tr}\mathbf{C}}$	Per-bin median rel. σ
ML noise (lgbm seedscan, $n = 10$)	5.061×10^{-41}	0.166%
Statistical (Poisson bootstrap, $N = 300$)	1.828×10^{-40}	0.564%
Systematic (matcorr sweep + norm)	2.463×10^{-39}	4.783%
Combined (block sum)	2.470×10^{-39}	4.822%
Paper <code>TotalCov</code> (reference)	2.676×10^{-39}	6.86%

The combined OmniFold envelope is $\approx 70\%$ of the paper’s per-bin median uncertainty, dominated by the systematic block. (The combined $\sqrt{\text{Tr}}$ is the quadrature sum of the three

independent component traces; the per-bin median 4.822% is taken directly from the matcorr rollup summary.)

4 Comparing to the paper

4.1 The 205-bin selection

The paper reports $d^2\sigma/(dp_T dp_{\parallel})$ on a 14-bin p_T axis $\times$ 16-bin $p_{\parallel}$ axis = 224 cells (edges in `minerva_paper_anc/bin_mapping.txt`). Of those, 205 have $x^{\text{paper}} > 0$ in `data_result_ptpl_2D_minerva.in`; the remaining 19 are paper-unreported cells that the paper’s analysis suppresses via an efficiency cut. We drop those 19 from both sides of every χ^2 and pull — that is the unambiguous “apples-to-apples” choice.

A secondary diagnostic mask, used only as a sensitivity check, is the strict-interior set: cells where $p_T^{\text{high}}/p_{\parallel}^{\text{low}} \leq \tan 20^\circ$. Some cells in the 205-bin set straddle the $\theta_\mu < 20^\circ$ kinematic cut and have wildly suppressed paper efficiencies that we do not replicate; pulls on those cells run to $\mathcal{O}(100)\sigma$ and dominate the χ^2 . The 185-bin interior cut removes them; we report it only for sensitivity. **The canonical metric is the 205-bin set** (commit 65fb085, *SeedsScan: use the 205 paper-reported bins everywhere*).

4.2 Covariance-aware χ^2/ndf

Math. With $\Delta \equiv \mathbf{x}^{\text{ours}} - \mathbf{x}^{\text{paper}}$ on the 205-bin set and $\mathbf{C}$ the chosen covariance,

$$\chi^2 = \Delta^\top \mathbf{C}^+ \Delta, \quad \text{ndf} = n_{\text{reported}} = 205, \quad \chi^2/\text{ndf} = \chi^2/205. \quad (20)$$

The implementation (`chi2_with_cov` in `compare_to_paper_fullcov.py`) builds the 205×205 block, inverts with `np.linalg.pinv` (§2.3), and forms the quadratic form.

Code pointer. `compare_to_paper_fullcov.py` (headline); `--omnifold-cov ROOT:HIST` (repeatable) adds each OmniFold covariance via Eq. (19); `--subtract-stat` removes the paper’s stat block to avoid double-counting the bootstrap (§5).

Worked example: comparison to paper (STATUS headline).

Metric (205 bins)	Paper	OmniFold (combined)
Width-weighted total σ , $\sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}}/(\mathbf{w}^\top \mathbf{x})$	4.61%	2.26%
Per-bin median relative σ	6.86%	4.822%
χ^2/ndf	3.661 (paper-only)	1.699 (paper+ours)
Log-normal χ^2/ndf	—	1.688

The $3.661 \rightarrow 1.699$ drop (-54%) is the central result of the UQ campaign: once the OmniFold-side error budget is honestly included, the agreement with the paper improves to $\chi^2/\text{ndf} \approx 1.7$. (A perfectly correct model with correct uncertainties gives exactly 1; 1.7 tolerates small residual shape mismatch in high- $p_{\parallel}$ tails, see §4.4.) **This combined χ^2 double-counts**

systematics shared with the paper (flux, GENIE, RPA, 2p2h, MINOS); see §5 for why it is still the honest headline and why an ours-only χ^2 is not yet quotable. The log-normal variant (matching the paper’s Table I treatment of the correlated flux normalization) gives 1.688, essentially identical.

Edge-bin sensitivity ($p_{\parallel}$ -min cut). `compare_to_paper_fullcov.py` ships a side-table that recomputes χ^2/ndf on progressively shrinking subsets, using the paper covariance only (no OmniFold add):

$p_{\parallel} \geq (\text{GeV}/c)$	N_{bins}	χ^2	χ^2/ndf	median ratio
1.5	205	750.49	3.661	1.0064
2.0	196	685.36	3.497	1.0059
2.5	186	545.87	2.935	1.0059
3.0	175	474.84	2.713	1.0059
3.5	163	452.95	2.779	1.0073
4.0	151	428.72	2.839	1.0068

The residual is concentrated in the low- $p_{\parallel}$, near-edge cells; the high- $p_{\parallel}$ tails carry sub-2% shape disagreement. This is a robustness check, not a separate method.

4.3 Pull distribution

Math. For each of the 205 reported bins,

$$\text{pull}_i = \frac{x_i^{\text{ours}} - x_i^{\text{paper}}}{\sqrt{C_{ii}^{\text{tot}}}}, \quad \mathbb{E}[\text{pull}_i] = 0, \quad \text{Var}[\text{pull}_i] = 1 \text{ under the null.} \quad (21)$$

Summary diagnostics: pull mean, RMS, the 14×16 pull heatmap, and a pull histogram (in `compare_to_paper_fullcov.py`).

Worked example (STATUS headline). With the combined covariance (paper TotalCov + matcorr universe + bootstrap-300 + ML):

- Pull mean: 0.069 (centred on zero; no global offset).
- Pull RMS: 0.466.

Pull RMS < 1 means the combined error band is *conservative* relative to the actual disagreement — the band is wider than it needs to be to absorb the observed scatter. No bins have $|\text{pull}| > 3$ in the reported set. Compare with paper-cov-only: mean 0.089, RMS 0.598 — similar mean (no bias), slightly wider spread before we include our own uncertainties.

4.4 Shape-only χ^2 with Jacobian propagation

What it estimates. Whether the *shape* of the $(p_T, p_{\parallel})$ cross section matches the paper, independent of absolute normalisation. Useful because the absolute normalisation depends on the flux integral, target nucleon count, and POT (all of which can drift by $\mathcal{O}(\%)$), whereas the shape is a pure unfolding diagnostic.

Definition. With bin widths $w_j = (\Delta p_T)_j (\Delta p_{\parallel})_j$ and total $T(\mathbf{x}) = \sum_{j \in \text{mask}} w_j x_j$, the unit-area shape is

$$s_i(\mathbf{x}) = \frac{x_i}{T(\mathbf{x})}. \quad (22)$$

By construction $\sum_i w_i s_i = 1$.

Jacobian. We need the covariance of $\mathbf{s}$ given the covariance of $\mathbf{x}$. Differentiating,

$$\frac{\partial s_i}{\partial x_j} = \frac{\delta_{ij}}{T} - \frac{x_i w_j}{T^2} = \frac{1}{T}(\delta_{ij} - s_i w_j). \quad (23)$$

In matrix form (in `normalize_xsec_shape.py`),

$$\mathbf{J} = \frac{1}{T}(\mathbf{I} - \mathbf{s} \mathbf{w}^\top), \quad \mathbf{C}^{\text{shape}} = \mathbf{J} \mathbf{C}^{\text{paper}} \mathbf{J}^\top. \quad (24)$$

Rank reduction. $\mathbf{J}$ is rank $n - 1$: the unit-area constraint $\sum_i w_i s_i = 1$ kills one degree of freedom. Algebraically, $\mathbf{w}^\top \mathbf{J} = \mathbf{w}^\top (\mathbf{I} - \mathbf{s} \mathbf{w}^\top)/T = (\mathbf{w}^\top - \mathbf{w}^\top)/T = \mathbf{0}^\top$, so the “all-bins-up” direction is the kernel of $\mathbf{J}$ and lives in the null space of $\mathbf{C}^{\text{shape}}$. Accordingly:

- $\mathbf{C}^{\text{shape}}$ has one zero eigenvalue and rank $n - 1$.
- $\chi_{\text{shape}}^2 = (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})^\top (\mathbf{C}^{\text{shape}})^+ (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})$ (pseudo-inverse drops the constrained direction).
- $\text{ndf} = n_{\text{bins}} - 1 = 204$ on the 205-bin set.

Code pointer. `normalize_xsec_shape.py`: `shape_jacobian`, `shape_chi2`.

Worked example (STATUS headline). Shape χ^2/ndf on 205 bins with the paper’s `TotalCovariance` (Jacobian-propagated): 3.596. This is essentially the absolute-normalisation χ^2/ndf of 3.661 *minus* the contribution of overall scale. Total- σ ratio is $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.0111$ — 1% scale difference. Shape disagreement at the $\sim 1\%$ level (median bin ratio 1.0064, 94% of bins within 10% of the paper) accounts for the remaining shape residual.

5 Ill-conditioning, rank deficiency, and open questions

The obstruction. For publication we would like an *ours-only* χ^2/ndf — the data-to-model agreement evaluated with our own covariance $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}}$ rather than the paper’s V . That requires inverting $\mathbf{C}^{\text{ours}}$, and it is rank-deficient:

$$\text{rank}(\mathbf{C}_{\text{sample}}^{\text{syst}}) = 140/205, \quad \text{cond}(\mathbf{C}^{\text{syst}}) = 8.7 \times 10^8. \quad (25)$$

The deficiency is set by **band count, not universe count**: under the MAT formula Eq. (16) each $\pm 1\sigma$ pair is rank 1 and the 100-universe PPF band is rank ≤ 99 , so the 44-band sum cannot fill the 205-dimensional bin space. *This is not a missing-bands problem.* The paper’s own systematic-only covariance (`TotalCov` – `StatOnlyCov`) is rank 120 — *lower* than our 140 — so we propagate at least as many independent systematic directions as

the published analysis (we use the same `GetStandardSystematics`). The paper’s V is full-rank (rank = 204, cond = 1.5×10^{12}) purely because it **includes the statistical block** (`StatOnlyCov`, rank 205, near-diagonal), not because it sums richer systematics. Adding our 300-replica bootstrap does the same for us: rank($\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$) = 201, cond = 3.8×10^{14} . (An earlier, lower-replica bootstrap left the rank near 140 — the origin of the now-superseded “bootstrap barely changes the picture” note.)

The real obstruction is statistical, not rank. Even at rank 201 a defensible single ours-only χ^2 is out of reach — but for a *statistical-precision* reason, not a missing-band one. Our OmniFold bootstrap stat block is $2.5\times$ smaller by trace than the paper’s stat block ($\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.8 \times 10^{-40}$ vs $\sqrt{\text{Tr}\mathbf{StatOnlyCov}} = 4.6 \times 10^{-40}$), so it regularises the small systematic modes $2.5\times$ less. The naive pseudo-inverse then gives $\chi^2/\text{ndf} = 252$ for $\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$, dominated by poorly-determined directions, against the paper’s 3.66 with its larger stat floor. The lever toward an ours-only χ^2 is therefore the *statistical* block — not the hunt for more systematic bands. The gap is *not* a missing MC-statistical term: our bootstrap already draws independent Poisson(1) multipliers on *both* data and MC per replica (§3.2; Practical Guide 2507.09582 §5.1), so data-stat and MC-stat contribute jointly. With that excluded, the smaller stat covariance most likely reflects genuinely higher unbinned-OmniFold statistical efficiency relative to the published binned D’Agostini unfold (whose iterative regularisation amplifies statistical fluctuations), or a difference in what the paper’s `StatOnlyCov` encompasses — the open physics question, and a more productive one than counting bands.

Regularization scans (all cutoff-dependent). On $\mathbf{C}^{\text{ours}}$ over the 205 reported bins, no regularizer gives a stable single number:

Method	Setting	Result
Eigenvalue truncation	$\lambda/\lambda_{\text{max}} > \{10^{-3}, 10^{-4}, 10^{-5}\}$	rank {19, 49, 82}; $\chi^2/205 = \{0.13, 1.12, 3.08\}$
Ridge $\mathbf{C} + \varepsilon\lambda_{\text{max}}\mathbf{I}$	$\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$	$\chi^2/205 = \{0.35, 1.38, 4.30\}$
Diagonal-target shrinkage	$\lambda = 0.05$	rank 178 $\rightarrow$ 204, cond 1.3×10^{11} , $\chi^2/205 = 3.2$

Each method spans roughly two decades of $\chi^2/205$ across a two-decade scan of its knob, with no defensible single cutoff. The useful conclusion is diagnostic, not a number: the naive inverse is dominated by tiny, poorly-determined modes, and in the retained high-variance modes the central-value difference is not large. **Quote direct per-bin pulls (§4.3) and this scan, not a single unqualified ours-only χ^2 .**

Bootstrap/paper-stat double-count. The combined headline (§4.2) adds $\mathbf{C}^{\text{boot}}$ on top of the paper’s `TotalCov`, which already contains the paper’s statistical block — a double count. `--subtract-stat` replaces the paper baseline by `TotalCov - StatOnlyCov` before adding our covariances, driving χ^2/ndf $1.699 \rightarrow 23.96$. That *over-corrects*: our $\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.83 \times 10^{-40}$ is far smaller than the paper’s published stat block ($\sim$ the 2.7×10^{-39} scale of `TotalCov`), so subtracting the latter without a comparable replacement removes too much. The headline

therefore keeps the double-counted 1.699 as the well-defined “what does OmniFold UQ add on top of the published V ” statement, with the de-double-counted number on file as a secondary diagnostic.

Open questions for MINERvA collaborators.

1. **Statistical block, not missing bands.** We previously read the 140/205 rank as missing systematic bands; direct comparison refutes this — the paper’s systematic-only covariance is rank 120 < 140, so our band set is at least as rich. The paper’s full rank is its statistical block, which we match with the bootstrap ($\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$: rank 201). The residual is that our bootstrap stat covariance is $2.5\times$ smaller by trace than the paper’s stat block (§5). This is not an MC-statistics omission — our bootstrap already Poisson-fluctuates both data and MC jointly. So either unbinned OmniFold is genuinely more statistically efficient than the binned D’Agostini unfold, or the paper’s **StatOnlyCov** encompasses more than a data+MC Poisson covariance. Which is it? Resolving this — not adding bands — is what would enable a defensible ours-only χ^2 .
2. **Bashyal flux $\leftrightarrow$ Muon_Energy_MINOS joint block** (JINST 16 P08068) — the only cross-band correlation the paper names. We *rederived* it: **MuonEnergyScale** is rank-1, so the cross term is rank-2, $\mathbf{C}_\times = \rho(\mathbf{d}_f \mathbf{d}_e^\top + \mathbf{d}_e \mathbf{d}_f^\top)$, with $\rho \approx 0.87\text{--}0.90$ recovered two independent ways from the shipped blocks (Cauchy–Schwarz in **TotalCov** = +0.90; subtraction residual = +0.87). It is real and now quantified, but adding it leaves our rank unchanged (its directions already lie in $\text{range}(\mathbf{C}^{\text{syst}})$). For a paper-independent ρ , is the fitted $(\Phi_{\text{norm}}, \Phi_{\text{shape}}, \text{MuonE}_{\text{scale}})$ posterior available from MAT-MINERvA?
3. **FrInel_pi.** It is commented out in **MAT-MINERvA/universes/GenieSystematics.cxx:37–38** with the note that “this knob should not currently be evaluated.” We are consistent with MAT in excluding it; the question is whether that exclusion is still current guidance.
4. **PPFX effective rank.** Our 100 PPFx universes appear to sample a much lower-dimensional nuisance model. Is there a recommended principal-component / fixed-rank flux covariance, rather than treating the 100 universes as independent samples?
5. **Publication precedent.** Have other MAT-based OmniFold / unbinned analyses hit this rank issue, and what did they publish for an ours-only χ^2 ?

What we do regardless. The publication-grade headline uses the combined covariance for $\chi^2/\text{ndf} = 1.699$ with the double-count caveat documented; the per-band rollup is MAT-conformant and byte-for-byte verified (§3.4); the 1.4% normalization is included as a rank-1 band; and the shrinkage-regularized ours-only cov is reported as a secondary characterization with λ disclosed. Adding the Bashyal joint block (and any per-target detector-mass shape) is the main outstanding item that would move us toward a defensible ours-only V .

6 Anatomy of the paper-covariance tension

The paradox. The central values agree well: $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.011$, median bin ratio 1.006, and the *per-bin pull RMS* against the paper covariance is only 0.598 (§4.3). A diagonal-only

χ^2 — ignoring all off-diagonal correlations — is therefore

$$\chi_{\text{diag}}^2/\text{ndf} = \frac{1}{n} \sum_i \frac{\Delta_i^2}{C_{ii}} = \text{RMS}(\text{pull})^2 + \text{mean}(\text{pull})^2 \approx 0.366, \quad (26)$$

yet the full-covariance $\chi^2/\text{ndf} = 3.661$. The tension is a $10\times$ inflation that lives *entirely* in the off-diagonal / small-eigenvalue structure of the published `TotalCov` (`diagnose_tension.py`). The rest of this section dissects it.

Eigenmode decomposition. Diagonalise the reported-block paper covariance, $\mathbf{C} = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ ($\lambda_1 \geq \dots \geq \lambda_n$), and project the residual $\mathbf{\Delta}$ onto each eigenvector, $c_k = \mathbf{u}_k^\top \mathbf{\Delta}$. Then

$$\chi^2 = \mathbf{\Delta}^\top \mathbf{C}^+ \mathbf{\Delta} = \sum_k \frac{c_k^2}{\lambda_k}, \quad \text{“eigen-pull”}_k \equiv \frac{c_k}{\sqrt{\lambda_k}}, \quad (27)$$

verified to reproduce the `pinv` $\chi^2 = 750.49$ to 3×10^{-9} . The decomposition shows the tension is **neither a few pathological null directions nor the absolute smallest eigenvalues**:

- The top contributing modes sit at $\lambda_k/\lambda_{\text{max}} \sim 10^{-4}$ – 10^{-5} — the lower-middle of the spectrum by magnitude, but still ~ 7 – 8 decades above the cond-floor ($\lambda/\lambda_{\text{max}} \sim 10^{-12}$) — with genuine eigen-pulls of 5 – 6σ .
- The ten *smallest*- λ modes carry only 3% of the χ^2 ; about 90 modes are needed to reach 90%. The χ^2 is broadly distributed, not localised to the ill-conditioned tail.

Regularisation robustness. A `pinv-rcond` scan confirms the tension is not a numerical artifact: retaining only the 73 best-determined directions ($\lambda_k/\lambda_{\text{max}} > 10^{-4}$) still gives $\chi^2/\text{ndf} = 1.42$; 139 directions ($> 10^{-6}$) give 2.79. A rank-truncation scan keeping the r largest- λ modes rises smoothly $0.69 (r=50) \rightarrow 2.35 (100) \rightarrow 3.30 (180) \rightarrow 3.66 (205)$ — no late jump, i.e. the tension does not ride on the last few least-determined modes.

It is shape, not normalisation. Splitting $\mathbf{\Delta}$ into a component along the measured spectrum $\mathbf{x}^{\text{paper}}$ (an overall scale) and the orthogonal remainder, the scale piece contributes $\chi_{\text{norm}}^2 = 0.10$ and the shape piece $\chi_{\text{shape}}^2 = 750.1$. The +1.1% normalisation offset is statistically irrelevant to the tension; it is a correlated *shape* difference. (Consistent with the Jacobian shape-only $\chi^2/\text{ndf} = 3.60$, §4.4.)

Where it lives. Mapping the dominant eigenvectors and the per-bin contribution $\Delta_i(\mathbf{C}^+ \mathbf{\Delta})_i$ back onto the 14×16 grid (`tension_mode_maps.png`, `tension_chi2_map.png`) localises the tension to the **low- p_T cross-section peak ridge** ($p_T \lesssim 0.4$ GeV/c, p_T -bins 2/7/10 carry 16/11/12%) and the low- $p_{\parallel}$ near-edge cells ($p_{\parallel}$ -bins 0/1/8). The carrier modes are sign-alternating *shape* oscillations across the peak, not vertical p_T -columns (which a flux $1/\Phi(p_T)$ normalisation error would produce) and not the $\theta_{\mu} < 20^\circ$ acceptance boundary. This is exactly where the flux and Muon.Energy bands — the two largest, and the only named cross-band correlation (Bashyal, §5) — dominate the covariance: tight bin-to-bin correlation there leaves little freedom to absorb a coherent ~ 1 – 2% shape difference, so a small OmniFold-vs-IBU shape difference becomes a large χ^2 .

A methodological component (estimator and iteration). Part of the tension is regularisation-dependent — the same data, same normalisation ($\Sigma\sigma$ within 0.1%), unfolded with a different GBDT backend moves the paper-cov χ^2 by ~ 1 unit while central values barely change (iteration count, by contrast, is negligible):

OmniFold configuration	paper-cov χ^2/ndf	pull RMS
exact-GBT, 5 iter (frozen production)	3.66	0.598
HistGBT, 5 iter (10-seed mean)	2.70 ± 0.04	0.56
LightGBM, 5 iter	2.65	0.563
LightGBM, 10 iter	2.66	0.565

Two things separate cleanly. (i) **Iteration count is negligible:** LightGBM 5 $\rightarrow$ 10 iters moves the χ^2 by $+0.01$ ($2.645 \rightarrow 2.657$), so OmniFold is well-converged at 5 iters and the tension is *not* an under-iteration artifact. (ii) **The GBDT estimator carries a ~ 1 -unit band:** the frozen exact-GBT production (3.66) is the outlier, while both histogram-binned backends agree at 2.65–2.70 (HistGBT is a 10-seed mean, range 2.63–2.74; the ± 0.04 ML-seed jitter is $\sim 25\times$ smaller than the 0.96 exact $\rightarrow$ hist gap, so that gap is genuinely the estimator, not seed noise). The exact-gradient trees fit finer shape structure (less implicit smoothing) and so deviate more from the paper’s IBU-regularised result. This is direct evidence that ~ 1 χ^2 -unit of the tension is OmniFold’s implicit regularisation rather than a physical data–MC disagreement (`tension_iter/`).

Conclusion. The tension is (i) a genuine, broadly-distributed, *correlated shape* difference — not a normalisation offset and not a small- λ inversion artifact — concentrated in the low- p_T peak where the flux/Muon_Energy bands dominate, plus (ii) a ~ 1 - χ^2 -unit methodological band from the GBDT estimator/iteration regularisation. Both point to the same next step: the missing Bashyal flux $\leftrightarrow$ Muon_Energy joint block (§5, open question 1) is exactly the covariance structure governing the region that carries the tension.

7 Calibration: coverage toys

What it estimates. Whether the bootstrap covariance band actually covers the truth at the rate it claims. A bootstrap band that is wrong — too tight or too loose — would give the right shape but the wrong χ^2 . Coverage toys provide an independent calibration check.

Procedure (`sbatch_coverage_toys_MEFHC_200.sh`). Generate $N_{\text{toy}} = 200$ closure unfolds, each one combining two existing modes of `unfold_2d_omnifold_unbinned.py`:

- `--closure`: pseudo-data are MC reco events (weighted), so the reference truth marginal $\mathbf{X}^{\text{truth}}$ is known exactly and is written to `hTruthXSec2D` in every output file.
- `--bootstrap-seed` k (seeds chosen to avoid collision with the production replicas): per-event Poisson(1) draws, same as the production bootstrap.

Each toy is independent.

Math. Stack the toys into an $N_{\text{toy}} \times 14 \times 16$ array $\mathbf{U}^{(t)}$. For each reported bin (where $\bar{X}_i^{\text{truth}} > 0$, i indexing the 205 reported cells):

$$\sigma_i = \text{std}(\mathbf{U}_i^{(t)}; \text{ddof} = 1) \quad (\text{spread across toys}), \quad (28)$$

$$r_i^{(t)} = \frac{U_i^{(t)} - \bar{X}_i^{\text{truth}}}{\sigma_i} \quad (\text{per-toy, per-bin standardised residual}), \quad (29)$$

$$c_i = \frac{1}{N_{\text{toy}}} \sum_{t=1}^{N_{\text{toy}}} \mathbf{1}\{|r_i^{(t)}| \leq 1\} \quad (\text{per-bin coverage}). \quad (30)$$

The expected value of c_i under a well-calibrated Gaussian band is the central- σ probability of a unit normal:

$$c^{\text{expected}} = \Phi(1) - \Phi(-1) = \text{erf}(1/\sqrt{2}) \approx 0.6827. \quad (31)$$

A second diagnostic is the mean absolute pull, $\langle |r| \rangle$, which for a unit normal is $\sqrt{2/\pi} \approx 0.798$.

Code pointer. `uq/coverage_toys.py` (rollup); driver `sbatch_coverage_toys_MEFHC_200.sh`; outputs `uq/coverage/2d_xsec_MEFHC_5iter_lgbm_coverage_toy*.root` and `uq/coverage/coverage_sum`

Worked example (STATUS, coverage). With $N_{\text{toy}} = 200$ on the 205 reported bins:

- Mean coverage: 68.71% (target 68.27%; $\Delta = +0.44\%$); median 68.50%.
- $\langle |r| \rangle = 0.794$ (target 0.798; within 0.5%).
- Signed residual mean: $+0.006 \pm 0.082 \sigma$ (no bias).
- Fraction of bins above the 65% target: 97.6% — only 5 of 205 bins fall below.

The agreement of $\langle c \rangle$ and $\langle |r| \rangle$ with their expected values at the half-percent level is the cleanest demonstration that the Poisson-bootstrap band is properly calibrated. The 200-toy ensemble (vs the Stage-1 20 toys) shrinks the per-bin binomial uncertainty to $\sqrt{0.68 \cdot 0.32/200} \approx 3.3\%$, which is why 97.6% of bins now clear the 65% target.

8 Closure tests: detecting bias

The variance methods of §3 answer the question “how much does the unfold wobble.” Closure tests answer a different question: “is the unfold biased — does it return a value that is consistently offset from the right answer when we inject a known truth?” Bias does not show up in the variance procedures; you have to look for it.

The three closure tests inject a known deformation into the MC, unfold, and check that the answer reproduces the deformation. Each test has its own “acceptance threshold,” set comfortably above the ML-noise floor of 0.166%.

8.1 Truth-reweight closure

Procedure. For a known shape $f(p_T, p_{\parallel})$, multiply both the MC reco weights *and* the MC truth weights by f at the event level. Unfold the reweighted pseudo-data against the unchanged (CV) response. The reference is the reweighted truth marginal — the unfolded answer should match it.

Two shapes tested.

- **gauss_pt**: $f = 1 + A \exp[-((p_T - p_T^0)/\sigma)^2]$ with $A = 0.2$, $p_T^0 = 0.4$ GeV/c, $\sigma = 0.1$ GeV/c.
- **tilt_pz**: $f = 1 + \alpha(p_{\parallel} - p_{\parallel}^{\text{ref}})$ with $\alpha = 0.1$, $p_{\parallel}^{\text{ref}} = 5$ GeV/c.

Acceptance. Median per-bin $|\text{residual}| \leq 1.5\%$.

Code pointer. `uq/closure/closure_truth_reweight.py`.

Worked example (1A lgbm 5-iter).

- **gauss_pt**: median $|\text{residual}| = \boxed{0.046\%}$ (PASS).
- **tilt_pz**: median $|\text{residual}| = \boxed{0.013\%}$ (PASS).

Both are $\sim 100\times$ tighter than the threshold; OmniFold recovers the injected truth shape essentially perfectly.

8.2 Hidden-variable closure

Procedure. This is the subtle one. The OmniFold feature set sees $(p_T, p_{\parallel})$ on reco and truth, but *not* the resolution variable $\Delta p_T \equiv p_T^{\text{reco}} - p_T^{\text{truth}}$. So we inject a bias on Δp_T only — a kinematic axis the network cannot see — and ask: how much of that bias leaks into the measured $(p_T, p_{\parallel})$ marginal?

The pseudo-data weight bump is

$$w^{\text{reco}} \rightarrow w^{\text{reco}} \cdot \left(1 + A \exp[-((\Delta p_T - c)/\sigma_{\Delta p_T})^2]\right), \quad (32)$$

with $c = +0.1$ GeV/c, $\sigma_{\Delta p_T} = 0.05$ GeV/c. Truth weights are *unchanged*, so the reference is the unaltered CV truth marginal. The amplitude is scanned: $A \in \{0.05, 0.10, 0.30\}$.

Leakage coefficient. Define the signed residual $\bar{r}(A)$ as the median of the per-bin signed residual at amplitude A . The leakage coefficient is the slope of a linear fit:

$$\bar{r}(A) \approx \kappa_{\text{leak}} A \quad \Rightarrow \quad \kappa_{\text{leak}} = \frac{\partial \bar{r}}{\partial A}. \quad (33)$$

κ_{leak} tells you what fraction of an unseen Δp_T -shaped bias contaminates the visible answer.

Acceptance. Median per-bin $|\text{residual}| \leq 3.0\%$ (looser than truth- reweight because Δp_T and p_T are weakly correlated, so the unfolding can never fully kill the leak).

Code pointer. `uq/closure/closure_hidden_var.py`.

Worked example (1A lgbm 5-iter, 205 reported bins).

- Per-amplitude median residuals: 0.57%/1.14%/3.43% for $A = 0.05/0.10/0.30$ — linear in A , as expected for small perturbations.
- Linear fit slope: $\kappa_{\text{leak}} = \boxed{0.113}$.

About 11% of a hidden Δp_T bias leaks into the measured $(p_T, p_{\parallel})$ marginal — under the 3% threshold for the largest realistic deformation ($A = 0.30$). Larger leakage would indicate that Δp_T ought to be added to the feature set; the present number is small enough to be safely ignored.

8.3 Alternative-model closure

Procedure. Train the OmniFold response on the CV weights. Build the pseudo-data from an alternative-universe reweighting (e.g. **MaCCQE:0** $+1\sigma$, or **Flux:50** PPFX index). The reference is the in-acceptance-extrapolated alt truth:

$$\tilde{X}_i^{\text{ref}} = X_i^{\text{CV,truth}} \cdot \frac{X_i^{\text{alt,in-acc}}}{X_i^{\text{CV,in-acc}}}. \quad (34)$$

This scales the CV truth marginal by the ratio of in-acceptance counts between alt and CV, isolating the unfolding’s response to a model-shape change from acceptance/efficiency effects. MaCCQE probes an interaction-model shape change; Flux probes flux reweighting — two distinct kinds of model dependence.

Acceptance. Median per-bin $|\text{residual}| \leq 2.0\%$.

Code pointer. `uq/closure/closure_alt_model.py`.

Worked example (1A lgbm 5-iter).

- **MaCCQE:0** ($+1\sigma$): median $|\text{residual}| = \boxed{0.068\%}$ (PASS); in-acceptance bias -1.68% folded out by the extrapolation.
- **Flux:50** (PPFX idx 50): median $|\text{residual}| = \boxed{0.376\%}$ (PASS); in-acceptance bias -7.32% .

Both pass with $\gg 5\times$ headroom. The unfolding is robust to model variations on the order of $\pm 1\sigma$ of GENIE and full PPFX flux universes.

9 How the pieces fit together

We can now read the `2D_OMNIFOLD_STUDY_STATUS.md` headline in one sentence: *across 205 paper-reported bins, the combined OmniFold + paper error budget gives $\chi^2/\text{ndf} = 1.699$ and pull mean/RMS 0.069/0.466, with a pull-RMS- < 1 band that has been independently checked at the 0.5% level by 200 coverage toys, on top of a 0.166% ML-noise floor that is more than an order of magnitude below the systematic envelope.*

Audit trail of the headline

Number	Section	Procedure
$\chi^2/\text{ndf} = 1.699$	§4.2	Eq. (6) with $\mathbf{C}^{\text{paper}} + \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{ML}}$
$\mathbf{C}^{\text{syst}}$	§3.3	187 universes, MAT-conformant Eq. (16) + norm
$\mathbf{C}^{\text{stat}}$	§3.2	Poisson bootstrap, 300 replicas, Eq. (15)
$\mathbf{C}^{\text{ML}}$	§3.1	lgbm seedscan, $n = 10$ trials
$\mathbf{C}^{\text{paper}}$	STATUS.md	Paper’s published <code>TotalCovariance</code>
Pull RMS = 0.466	§4.3	Per-bin standardised residual on combined $\mathbf{C}$
Combined median rel. = 4.822%	§3.5	$\sqrt{\text{diag}\mathbf{C}^{\text{tot}}}/\mathbf{X}^{\text{CV}}$
Mean coverage = 68.71%	§7	Coverage toys, $N_{\text{toy}} = 200$

Standing assumptions

1. **Gaussian approximation.** Pulls and χ^2 treat each bin as Gaussian. Justified because per-bin entries are weighted sums over $\mathcal{O}(10^4)$ MC events; central limit theorem applies.
2. **Independence of error sources.** Eq. (19) requires no cross-correlations between ML, statistical, and systematic. The bootstrap pins the GBDT seed (so ML noise cancels at first order); universes pin the seed too; the seedscan pins the bootstrap. Cross-terms are higher-order. (The available $N = 300$ bootstrap block carries a small residual ML leak, §3.2.)
3. **Bootstrap calibration.** §7’s mean coverage = 68.71% certifies $\mathbf{C}^{\text{stat}}$ to half a percent.
4. **Band independence in $\mathbf{C}^{\text{syst}}$.** Different physics sources are taken as uncorrelated; the one known exception, the Bashyal flux $\leftrightarrow$ muon-E joint block, is not yet propagated (§5).
5. **Combined- χ^2 double-count.** The headline shares systematics with the paper’s V and is the “what OmniFold UQ adds on top of V ” statement, not an independent ours-only test (§5).
6. **Closure passes are valid for the perturbations tested.** Nothing guarantees they pass for unphysical deformations far outside the training distribution.

A Symbol table

Symbol	Meaning
$\mathbf{x}, \mathbf{X}$	205-vector of per-bin cross sections
$\mathbf{C}$	205×205 covariance matrix
Δ	$\mathbf{x}^{\text{ours}} - \mathbf{x}^{\text{paper}}$ or $\mathbf{X}^{(u)} - \mathbf{X}^{\text{CV}}$ depending on context
$\langle \mathbf{X} \rangle_B$	universe mean within band B (MAT reference)
$\mathbf{J}$	Jacobian of a transformation, Eq. (12)
N	number of bootstrap/seedscan replicas; N_u universes in a band
n	number of bins ($n = 205$)
b_i	Poisson(1) per-event bootstrap weight
s_i	unit-area shape, $s_i = x_i / (\mathbf{w}^\top \mathbf{x})$
w_i	bin width $(\Delta p_T)_i$ $(\Delta p_{\parallel})_i$ in the shape definition
$r_i^{(t)}$	per-toy, per-bin standardised residual (§7)
pull_i	per-bin standardised residual vs. paper (§4.3)
Φ, erf	standard normal CDF, error function

B Reproducing the numbers

```
# ML seedscan rollup (n=10, lgbm)
python3 seedscan/analyze_seedscan.py

# Bootstrap covariance (300 toys, MEFHC, lgbm)
python3 uq/analyze_uq.py \
    --glob "uq/2d_xsec_MEFHC_5iter_lgbm_boot*.root" \
    --outdir uq/bootstrap_MEFHC_300/ \
    --out-root uq_covariance_boot300.root

# Systematic covariance (187 universes, MAT-conformant + 1.4% norm)
python3 uq/analyze_universes.py \
    --cv uq/2d_xsec_MEFHC_5iter_lgbm_cv_full.root \
    --glob "uq/2d_xsec_MEFHC_5iter_lgbm_uni*.root" \
    --add-norm 0.014 \
    --outdir uq/universe_stage2_MEFHC_full_matcorr/ \
    --out-root uq_universe_covariance_full_matcorr.root

# MAT parity check (our rollup == Mnvh1D::GetTotalErrorMatrix)
python3 uq/verify_matcorr_vs_mnvh1d.py

# Headline  $\chi^2/\text{ndf}$  and pulls, combined cov
python3 compare_to_paper_fullcov.py \
    --omnifold-cov uq/universe_stage2_MEFHC_full_matcorr/ \
uq_universe_covariance_full_matcorr.root:hCov_universe_total \
    --omnifold-cov uq/bootstrap_MEFHC_300/uq_covariance_boot300.root:hCov2D_reported \
    --omnifold-cov uq/seedscan_lgbm_ml/uq_covariance_ml.root:hCov2D_reported
```

```
# Shape-only chi^2 with Jacobian
python3 normalize_xsec_shape.py

# Coverage toys (200)
python3 uq/coverage_toys.py \
    --toy-glob "uq/coverage/2d_xsec_MEFHC_5iter_lgbm_coverage_toy*.root" \
    --out-summary uq/coverage/coverage_summary.txt
```

C Multinomial $\rightarrow$ independent Poisson bootstrap limit

A non-parametric bootstrap with replacement draws N events from the original N -event sample. Let n_i be the number of times event i is drawn. Then $(n_1, \dots, n_N) \sim \text{Multinomial}(N, \mathbf{p})$ with $p_i = 1/N$ uniform, $\sum_i n_i = N$.

The marginal $n_i \sim \text{Binomial}(N, 1/N)$, whose mean is 1 and variance $(N-1)/N \rightarrow 1$. For large N , the binomial converges to Poisson(1) (Poisson limit theorem: many trials, small success probability, fixed expected count).

The full multinomial has the constraint $\sum n_i = N$, which introduces $\mathcal{O}(1/N)$ correlations between the n_i . In the large- N limit those correlations vanish and the n_i become *independent* Poisson(1) variables. This is the Poissonisation of the multinomial: the conditioned N -event multinomial is a Poisson process *conditioned* on hitting N counts, and the unconditional version is independent Poissons.

In practice, the independent-Poisson form is what we implement (one `np.random.default_rng().poisson` draw per event), and the $\sum n_i = N$ constraint is dropped. For $N \gtrsim 10^6$ (our regime), the relative error from the dropped constraint is $\mathcal{O}(1/\sqrt{N})$ on the total event count — far smaller than the per-bin Poisson variance.

D The $\pm 1\sigma$ pair covariance estimator: MAT vs legacy

For a single universe band with two universes $u = 0, 1$ at $\pm 1\sigma$, write the two value vectors as $\mathbf{X}_0$ and $\mathbf{X}_1$, with universe mean $\langle \mathbf{X} \rangle = \frac{1}{2}(\mathbf{X}_0 + \mathbf{X}_1)$. The **MAT-conformant** estimator Eq. (16) at $N_u = 2$ is

$$\mathbf{C}_{\text{MAT}}^B = \frac{1}{2} \sum_{u=0,1} (\mathbf{X}_u - \langle \mathbf{X} \rangle)(\mathbf{X}_u - \langle \mathbf{X} \rangle)^\top = \frac{1}{4}(\mathbf{X}_0 - \mathbf{X}_1)(\mathbf{X}_0 - \mathbf{X}_1)^\top, \quad (35)$$

a rank-1 outer product of the *half-difference*. It is mean-centered, so any common offset of the pair away from X_{CV} contributes nothing.

Contrast the **legacy** CV-centered pair estimator that the pre-2026-05-28 rollup used, with $\Delta_u = \mathbf{X}_u - \mathbf{X}_{\text{CV}}$ and decomposition $\Delta_0 = \mathbf{m} + \mathbf{a}$, $\Delta_1 = \mathbf{m} - \mathbf{a}$:

$$\mathbf{C}_{\text{legacy}}^B = \frac{1}{2}(\Delta_0 \Delta_0^\top + \Delta_1 \Delta_1^\top) = \mathbf{m} \mathbf{m}^\top + \mathbf{a} \mathbf{a}^\top. \quad (36)$$

The two agree *only* for a CV-symmetric pair ($\mathbf{m} = 0$), where both reduce to $\mathbf{a} \mathbf{a}^\top$. For an asymmetric or one-sided pair the legacy form adds a spurious rank-1 $\mathbf{m} \mathbf{m}^\top$ term (treating

the common offset as a second variance direction), which MAT correctly collapses to near zero. This is why moving to the MAT formula *lowered* the total universe trace by 1.9% (the BeamAngle, MuonResolution, and one-sided NC bands had asymmetric shifts), and it is the single methodological correction behind the χ^2/ndf shift $1.473 \rightarrow 1.699$. The legacy estimator remains available via `--legacy-pair-formula` for forensic reproduction only; the earlier claim that the CV-centered convention is “consistent with the published MINERvA error budget” was incorrect — MAT is mean-centered (§3.4).